

WATER QUALITY REQUIREMENTS FOR FIRST-FEEDING IN MARINE FISH LARVAE. II. pH, OXYGEN, AND CARBON DIOXIDE

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Abstract: Previously unfed larvae representing eight species of marine teleosts (a soleid, *Heteromyceteris capensis* Kaup; a cynoglossid, *Trulla capensis* Kaup; a gadid, *Gaidropsarus capensis* (Kaup); a congio-podid, *Congiopodus spinifer* (Smith); three sparids, *Diplodus sargus* (L.), *Pachymetopon blochi* (Val.), and an unidentified species; and a centracanthid, *Pterosmaris axillaris* (Boulenger)) were exposed for 24 h to various pH's and concentrations of dissolved oxygen and carbon dioxide and then tested for their ability to first-feed. Mortalities recorded during the 24-h exposure went into the fitting of response curves and the calculation of the LC_{50} and LC_{10} . The incidence of successful first-feeding during the subsequent 4-h exposure to food yielded values for the EC_{50} and EC_{10} . The exposure to sub-optimal conditions continued during the 4-h feeding test. The sensitivity of first-feeding larvae to high pH and dissolved oxygen may pose hazards for the marine fish culturist; low pH appears to be harmful (at least in the short term) only at levels below about pH 6.0; carbon dioxide poisoning is of doubtful importance in practical fish culture. There were only minor interspecific differences in calculated 24-h first-feeding EC_{50} 's, which suggests general applicability of the results to a wide variety of first-feeding marine fish larvae.

INTRODUCTION

The kinds of environmental stressors which, at present, are of particular interest to the intensive mariculture manager or engineer are those that originate with the cultured organisms themselves, or with their food. It is these stressors that determine water re-use rates, stocking densities, and other factors relating to water quality management. Low and high pH, low dissolved oxygen, and high carbon dioxide are examined in this study with the object of identifying and quantifying those that are likely to cause concern in a marine fish hatchery/nursery operation. Ammonia, nitrite, and nitrate were examined previously (Brownell, 1980).

The decline in the pH of recirculated water is chiefly attributable to the bacterial oxidation of excretory ammonia. From the stoichiometries of the reactions involved, Wickins (1976) calculated that the oxidation of 1 mg of nitrogen as NH_4^+ to nitrogen as NO_3^- requires 4.27 mg O_2 and 0.09 mg inorganic C, and produces 0.14 mg H^+ . The decrease in pH is compounded by a simultaneous loss of carbonate alkalinity and consequent decline in the buffering capacity of the recirculated water.

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The sensitivity of freshwater fish to low pH has been extensively studied (Doudoroff & Katz, 1950; EIFAC, 1968). Requirements for survival through egg and larval stages were still poorly known at the time of the EIFAC review. This gap has since largely been filled for freshwater fish.

The pH of oceanic surface waters in equilibrium with atmospheric carbon dioxide rarely falls below 8.1; the extreme of pH 7.5 is generally restricted to oxygen minimum zones, usually at depth (Sverdrup *et al.*, 1942). Without evidence to the contrary, one could easily accept the notion (Jones, 1964, p. 107) that marine fish are markedly less tolerant of low pH than freshwater fish. Although the response of certain marine fish (including early stages) to low pH has been studied (e.g., Bishai, 1960) it is not possible, from the published reports, to establish whether the observed effects were due to hydrogen ion concentration or to free carbon dioxide produced by the acidification of sea water.

Several investigators have published support for the increasingly widespread practice of rearing the early stages of marine fish in the presence of one or more species of planktonic algae (e.g., Alderson & Howell, 1973). The observed enhancement of survival and growth of the larvae has been attributed to various mechanisms (Struhsaker *et al.*, 1973), among them that phytoplankton provides food for the herbivorous zooplankters on which the fish are fed; that it extracts nitrogenous metabolites, e.g., ammonia, that could otherwise reach harmful levels; that it releases anti-bacterial, or even growth-enhancing substances into the rearing water. The effect of phytoplankton growth of interest here is the increase in pH that is brought about indirectly by the extraction of free carbon dioxide from the water.

Publications on the tolerance of freshwater fish to high pH have been reviewed by Doudoroff & Katz (1950), Jones (1964), and EIFAC (1968), among others. The literature includes several reports on the tolerance of eggs and larvae of both freshwater and marine species (e.g., Schach, 1939; Kelley, 1946; Bishai, 1960).

Reviews of the literature on the dissolved oxygen (DO) requirements of fish have been compiled by Doudoroff & Shumway (1970) and Davis (1975), among others. Marine fish have received only a fraction of the attention devoted to freshwater species. Available data on the response of marine fish larvae to low DO come almost entirely from Bishai (1959), Deubler & Posner (1963), Saksena & Joseph (1972), De Silva & Tytler (1973), and Sylvester *et al.* (1975). None of these studies was concerned with the effect of reduced DO on food intake.

From the literature on the DO requirements of various stages in the life cycles of teleost fish, it can be concluded (though not entirely without reservation) that the period beginning just prior to hatching and ending with the development of branchial respiration is the most sensitive to low DO. Such a period is present in most marine and freshwater teleosts with small eggs. There are several indirect reasons to suspect that marine fish larvae relying on cutaneous respiration would be particularly sensitive to low DO, and that this sensitivity would be aggravated in culture. (1) It is during the early feeding stages that maximal density of food is

required, which contributes to the biological oxygen demand of the rearing water. (2) The turbulence and shearing forces arising from direct aeration or spraying of tank water can be deleterious during the prefeeding and early feeding stages. (3) Although sea water can be more efficiently aerated than fresh water (Kils, 1977), the DO content of air-saturated sea water is about 20% less than that of air-saturated fresh water at the same temperature. (4) The toxicity of several substances, including ammonia, may be increased by decreases in DO (e.g., Lloyd, 1961). (5) Certain of the compensatory mechanisms available to fish with branchial respiration and haemoglobin, i.e., increasing opercular frequency and amplitude, and haemoconcentration, are obviously not possible during the phase of cutaneous respiration. Increased swimming activity or pectoral fanning might achieve the required ventilation, but at the cost of a higher metabolic rate.

The connection between low DO and loss of appetite reported by Stewart *et al.* (1967), among others, coupled with the above considerations, supports the proposition of Iwai & Hughes (1977) that the maintenance of adequate DO might be a very critical requirement for first-feeding success in marine fish larvae.

The literature on the effects of carbon dioxide on fish physiology and survival is not extensive; it has been reviewed by Doudoroff & Katz (1950), Fry (1957), and Jones (1964). In general, the reviewers conclude that fish are very tolerant of elevated concentrations of free carbon dioxide, particularly when they are allowed sufficient time to acclimate. Although the ability of fish to extract oxygen from surrounding water diminishes at abnormally high concentrations (the Root effect), the "ordinary environmental levels of carbon dioxide have little influence on the rate of oxygen consumption" in all but the "most sensitive species" (Fry, 1957). Fry stressed that considerable work needed to be done on the subject.

Aside from Kelley's (1946) experiment with herring (*Clupea harengus pallasii*) eggs, there appears to have been no attempt to determine the tolerance of marine fish eggs or larvae to elevated free carbon dioxide. The wide variation in carbon dioxide concentration during her experiment renders the results very difficult to interpret.

MATERIALS AND METHODS

The general procedures follow those described by Brownell (1980). Dilution water used in all experiments (hereafter referred to as "system water") came from the closed-circulation system described in that publication. Specific methods pertaining to certain of the variables tested are treated separately below.

TOLERANCE STUDIES

LOW pH

Specific methods

Quantities of a concentrated (≈ 4 N) HCl stock solution were micro-pipetted into experimental feeding containers (see Brownell, 1980) holding system water, and the mixture was vigorously aerated for 1 h before measuring pH (NBS). Sixty minutes of aeration was found to be sufficient to reduce free carbon dioxide concentration to within 10% of equilibrium levels. If the resulting pH was not satisfactory, additional HCl or additional system water was added and aeration was resumed. The larvae were first transferred from the larva baskets into system water which had been acidified to about pH 3, aerated, and brought to neutrality by addition of sodium hydroxide. From there, they were pipetted into the test solutions. Food was re-suspended in a small quantity of the decarbonated water prior to addition. A second pH reading was taken at the end, after the incidence of feeding had been recorded and the larvae removed. The nominal pH was the arithmetic mean of the initial and final pH's recorded. Uncertainty as to the actual pH during exposure was greatest near the pH inflection point ($\approx$ pH 4.5). The computed 95% confidence limits were expanded by empirically derived values that took pH uncertainties into account. A normal distribution of tolerance over units of pH was assumed in the fitting of response curves.

Results and discussion

Three species were tested: a gadid, a soleid, and a sparid. Only the response curves for the gadid are presented here in graphical form (Fig. 1). For all three

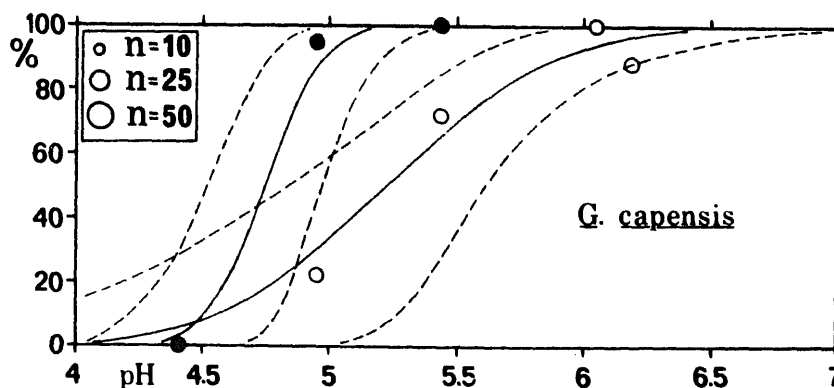


Fig. 1. Response of first-feeding *Gaidropsarus capensis* to low pH: open circles represent raw data (adjusted for natural responsiveness) on first-feeding incidence following 24-h exposure to various low pH's; solid circles represent data on percent survival during 24-h exposure; the areas of the circles are proportional to the number of larvae tested at corresponding pH's; the 95% confidence limits are represented by dashed lines.

TABLE I

Summary of tolerance limits: oxygen is expressed in $\text{mg} \cdot \text{l}^{-1}$, carbon dioxide in $\text{mg} \cdot \text{l}^{-1} \text{CO}_2\text{-C}$; the $\text{LC}_{50}/\text{EC}_{50}$ ratios supplied for low and high pH are based on hydrogen and hydroxyl ion activities, respectively, and not on units of pH; 95% confidence limits are in parentheses.

Variable	Species	24-h first-feeding EC_{50}	24-h first-feeding EC_{10}	24-h LC_{50}	24-h EC_{10}	$\text{LC}_{50}/\text{EC}_{50}$
Low pH	<i>Gaidropsarus capensis</i> (Kaup)	5.22 (4.84–5.61)	5.91 (5.54–6.27)	4.74 (4.52–4.97)	4.99 (4.77–5.21)	3.0
	<i>Heteromycteris capensis</i> Kaup	5.25 (4.97–5.53)	5.90 (5.52–6.29)	4.48 (4.31–4.65)	4.51 (4.33–4.69)	5.9
	<i>Diplodus sargus</i> (L.)	5.20 (4.93–5.47)	5.69 (5.42–5.96)	5.06 (4.84–5.27)	5.17 (4.94–5.39)	1.4
	<i>Gaidropsarus capensis</i>	8.67 (8.56–8.77)	8.39 (8.25–8.53)	9.24 (9.12–9.36)	8.99 (8.87–9.10)	3.7
High pH	<i>Heteromycteris capensis</i>	8.78 (8.64–8.92)	8.50 (8.29–8.71)	9.21 (9.06–9.35)	8.92 (8.79–9.06)	2.7
	<i>Diplodus sargus</i>	8.82 (8.69–8.95)	8.52 (8.34–8.70)	9.04 (8.84–9.14)	8.89 (8.78–9.00)	1.7
	<i>Pterommaris axillaris</i> (Boulenger)	4.77 (4.06–5.48)	5.66 (4.43–6.89)	3.49 (2.99–3.99)	3.98 (3.35–4.61)	0.73
	<i>Pachymetopon blochi</i> (Val.)	3.98 (3.24–4.71)	5.22 (4.12–6.31)	3.16 (2.67–3.66)	3.46 (2.89–4.02)	0.79
O_2 (static)	Species 5 (Sparidae)			3.56 (2.92–4.20)	3.94 (3.15–4.73)	
	<i>Trulla capensis</i> Kaup.	4.30 (3.58–4.02)	5.41 (4.49–6.33)	<2.32		<0.54
	<i>Gaidropsarus capensis</i>	4.01 (3.26–4.76)	5.62 (4.56–6.67)	2.50 (1.78–3.23)	2.92 (2.25–3.60)	0.62
	<i>Congiopodus spinifer</i> (Smith)	4.04 (2.90–5.19)	5.30 (4.15–6.45)	3.63 (2.98–4.27)	4.21 (3.48–4.94)	0.90
O_2 (flow-through)	<i>Gaidropsarus capensis</i>	3.15 (2.43–3.86)	4.06 (3.15–4.97)	2.06 (1.58–2.55)	2.73 (2.08–3.38)	0.65
	<i>Heteromycteris capensis</i>	3.09 (2.26–3.93)	4.03 (2.29–5.08)	0.94 (0.55–1.32)	1.18 (0.80–1.57)	0.30
CO_2	<i>Heteromycteris capensis</i>	>7.7				

species (Table I), the 24-h first-feeding EC_{50} 's fell between pH 5.2 and 5.3; the 24-h LC_{50} 's, between pH 4.5 and 5.1. First-feeding began to be affected only as pH was reduced below about 6.0. The tolerance of these larvae came as a surprise. The data show that marine fish are about as tolerant of low pH as freshwater species, at least in the short term. Experiments on eggs and larvae of various freshwater species have yielded tolerance limits (variously defined) at pH's below 6.0 (e.g., Johansson & Kihlström, 1975; Milbrink & Johansson, 1975; Trojnar, 1977a, b). In the publications just cited, test solutions were allowed to equilibrate with atmospheric carbon dioxide prior to exposure. Almost all studies on freshwater fish eggs or larvae where free carbon dioxide concentration may have been large enough to affect the results yielded higher pH's as limiting values. No published information on the tolerance of marine fish eggs or larvae to low pH was encountered where a satisfactory distinction was made between toxicity due to hydrogen ion and that due to carbon dioxide.

From the results of the present study, it might be asserted that the maintenance of pH and carbonate alkalinity in closed and semi-closed fish culture systems at levels approximating those in the sea is unnecessary. This conclusion was reached by Hirayama (1970) who monitored pH and alkalinity (as well as ammonia and nitrate) in a small closed marine system in which juvenile sparids (*Acanthopagrus*) were kept. Alkalinity decreased linearly with time to a value of $0.2 \text{ meq} \cdot \text{l}^{-1}$ and pH below 6.0 with the fish showing no abnormal symptoms. Although a pH-related inhibition of nitrification has been reported, the nitrifying capacity of Hirayama's filter was still adequate to keep ammonia at a low level. The reward of operating a recirculating culture system at a lower pH would be the reduction in $\text{NH}_3/\text{NH}_4^+$ ratio. Before any such recommendation can be made, however, it will first be necessary to weigh the gain in ammonia ratio against possible long-term pH effects on fish larvae, bacteria, and ionic balance of the recirculating water.

HIGH pH

Specific methods

Varying amounts of a strong ($\approx 4 \text{ N}$) stock solution of NaOH in distilled water were added to system water in 350-ml test containers to yield an array of treatment concentrations. The solutions were then aerated for an hour, pH (NBS) was measured, and larvae were added. The pH of the water was again measured after the larvae had been removed. Change in pH during the 28-h experiments was caused primarily by addition of system water along with the fish and food (10 and 5 ml, respectively). The nominal pH was the mean of the initial and final pH's recorded. The 95% confidence limits of computed response curves were expanded by addition or subtraction of empirically determined values that took into consideration the instrument error in measuring pH and the variation in pH during tests. A model assuming a normal distribution of tolerance over units of pH was employed in the fitting of the response curves.

Results and discussion

Tolerance limits among the three species tested (a gadid, a soleid, and a sparid) were very similar: a 50% reduction in first-feeding incidence occurred at $\approx \text{pH } 8.7$, 8.8 , and 8.8 in the gadid, soleid, and sparid, respectively (Table I). A 10% reduction occurred at $\approx \text{pH } 8.4$, 8.5 , 8.5 , respectively. 24-h LC_{50} 's obtained for the three species fell between 9.0 and 9.2. Based on inspection of the three response curves (one of which is presented here, Fig. 2), a negligible-effect pH is not likely to exceed pH 8.4. This value is only marginally higher than the more extreme pH's recorded in the open sea (pH 8.3: Sverdrup *et al.*, 1942), and in light of the trend toward rearing marine fish larvae in green water, might pose a serious limit for the nurseryman.

Comparisons of these values with those of previous workers is difficult. Schach (1939) reported the limits of tolerance for herring (*Clupea harengus*) larvae as pH 8.45, and the optimal range as 8.2–8.3, but failed to explain the source of these values. Bishai's (1960) herring larvae died when the pH was raised from 8.1 to 8.5 by addition of NaHCO_3 . The number of larvae tested and the final concentration of bicarbonate was not reported. Kelley (1946) found that Pacific herring eggs hatched a day sooner when they were incubated at pH 8.68 (range during exposure: 7.93–8.96), but that percent survival through hatching was not different from controls. A marked decrease in hatchability occurred in another experiment when pH rose to 9.12. No other data on the tolerance of the early stages of marine fish to elevated pH appear to be available.

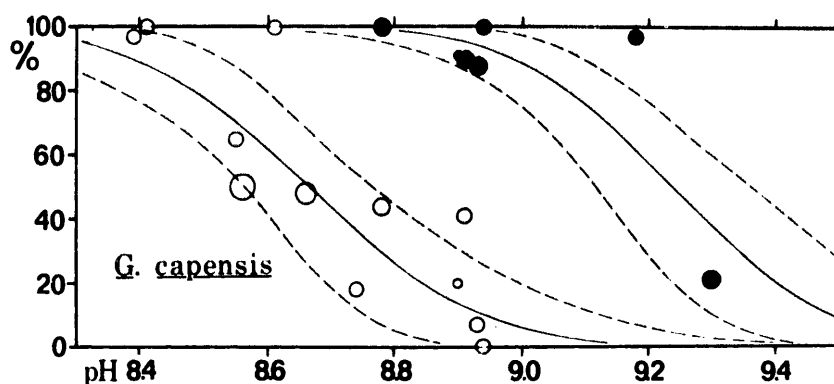


Fig. 2. Response of first-feeding *Gaidropsarus capensis* to high pH: other features as in Fig. 1.

A certain amount of acclimation to elevated pH has been demonstrated in juvenile rainbow trout by Jordan & Lloyd (1964). The improved growth rate of juvenile sole (*Solea solea*) over a 13-day period in tanks where the pH averaged ≈ 8.4 and rose eventually as high as 8.8 (due to phytoplankton production) (Alderson & Howell, 1973) suggest either that acclimation to elevated pH occurs among juvenile marine fish, or that they are simply more tolerant than larvae. It may be that the marine larvae in the present study, given a more gradual exposure increase, would have better tolerated high pH's. Further research on the response of fish larvae, both freshwater and marine, to elevated pH (particularly that associated with phytoplankton blooms) is needed.

DISSOLVED OXYGEN

Specific methods: static tests

System water, de-oxygenated to varying degrees by nitrogen stripping, was placed in test containers. Oxygen concentration was measured with a YSI-57 oxygen meter and probe. Larvae were introduced by pipette (along with 10 ml of system water) and the container was closed with a screened bung into which a small wad of synthetic wool had been pressed. After 24 h, the wool plug was removed, food

was added, and the plug was replaced in the bung. Oxygen was again measured immediately after the larvae had been rated for gut-fullness and removed.

Increases in the oxygen concentration during the course of the experiments came as a result of diffusion of atmospheric oxygen through the wool plug, by addition of oxygen with larvae and food, and during removal of larvae; decreases resulted from respiration by bacteria, food, and fish. A relationship of the oxygen concentrations measured at the beginning and end to the average concentration to which larvae were actually exposed was derived by fitting curves to measurements made during dummy runs. This permitted the calculation of an estimate of this average concentration for use as a nominal value in fitting response curves. Upper and lower 95% confidence limits were expanded by addition and subtraction, respectively, of empirically determined values to account for instrument error and variation in oxygen during exposures.

Specific methods: flow-through tests

The flow-through apparatus (Fig. 3) accommodates one control and four treatments. Water from the closed circulation system is de-oxygenated to varying degrees in a nitrogen stripping column, passed through the test containers holding the larvae, re-oxygenated, and returned to the system. Flow rate through each test container was maintained at $20 \pm 2 \text{ ml} \cdot \text{min}^{-1}$, which was sufficient to exchange 90% of the water in the containers every $40 \pm 3 \text{ min}$. This gravity-feed system, similar to that described by Granmo & Kollberg (1972) was found to be very reliable. Variation in flow rate due to bubble formation in the tubing was not a problem as long as the head was $>20 \text{ cm}$. For additional assurance, a $150 \text{ ml} \cdot \text{min}^{-1}$ diaphragm pump (Fig. 3) was used to dampen the effect of any variation in flow through one or more of the treatment channels.

At the end of the 24-h exposure, a count of survivors was made visually, without disconnecting the containers. A concentrated suspension of rotifers was then equally distributed among the five bubble traps. After 4 h exposure to food, one container at a time was disconnected, replaced by a container full of system water only, and then examined in the usual manner with the suspended microscope. In this way, the oxygen levels in the remaining containers were not affected by a decrease in the flow-rate of water through the column while the feeding incidence was being determined. Confidence limits of computed response curves were adjusted to account for instrument error and variation in oxygen concentration during the test. For both static and flow-through data, response rate was linearized by assuming a normal distribution of tolerance over un-transformed units of DO concentration.

Results and discussion

Results are presented for six species tested in static water and two species with the flow-through apparatus (Table I). Response curves for one species are pre-

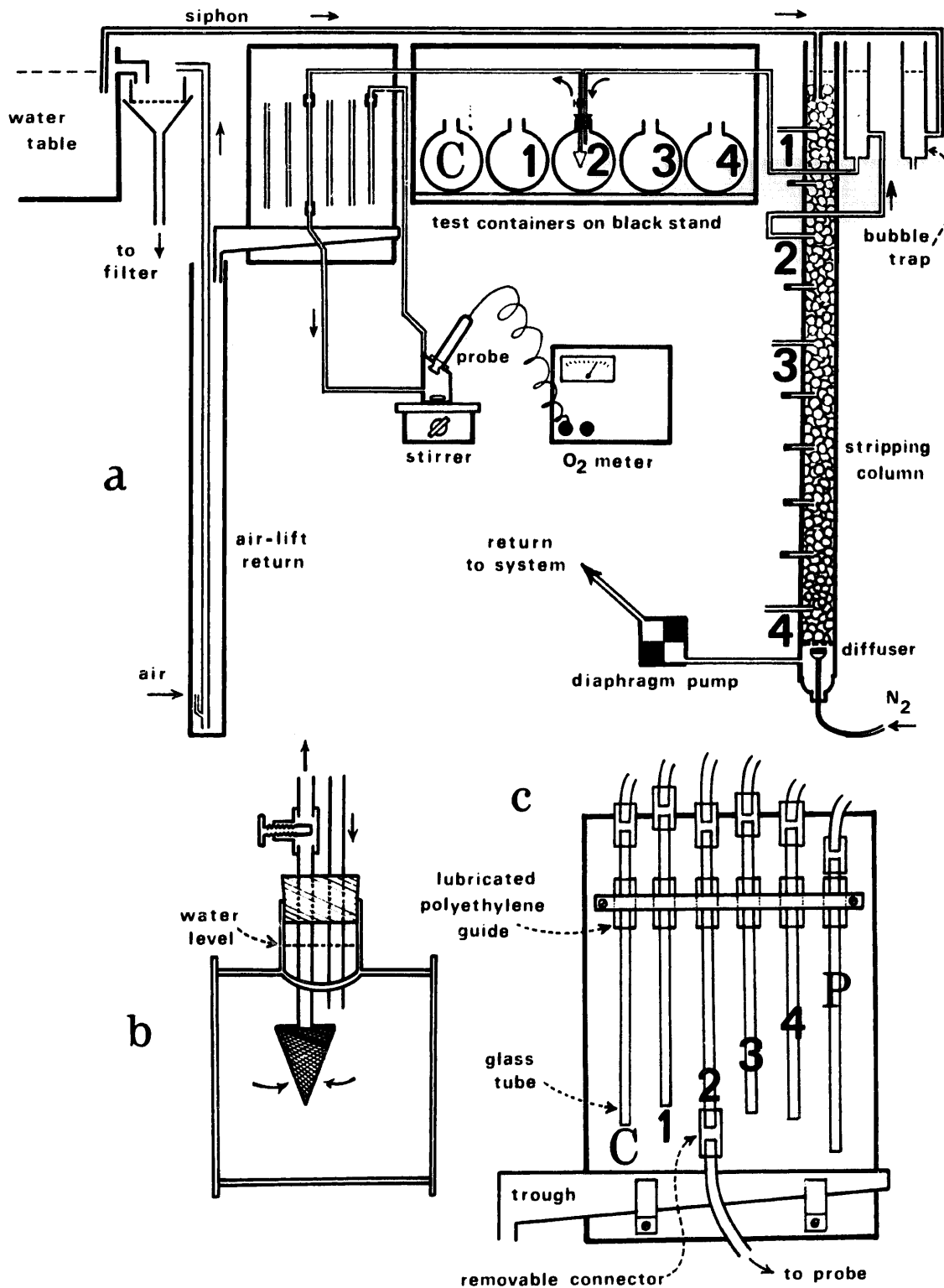


Fig. 3. The flow-through apparatus used for testing the tolerance of first-feeding larvae to low dissolved oxygen: a, general view, not to scale, only one channel (treatment No. 2) is shown, with probe circuit attached, two of the five bubble traps are shown; b, side view of test container with conical 40- μ m mesh and valve on outlet; c, flow-control device, rate of dripping was regulated by raising or lowering the glass tubes.

sented (Figs. 4, 5). First-feeding incidence in all species was reduced by 50%, in a static test, when the DO concentration of the water mass was between ≈ 4.0 and $4.75 \text{ mg} \cdot \text{l}^{-1}$. Caution must be exercised in predicting the response at more extreme percentiles. Although calculated EC_{10} 's (Table I) would suggest a somewhat higher value, the raw data show that the DO concentration at which first-feeding "begins" to be affected falls somewhere between 5.0 and $5.5 \text{ mg} \cdot \text{l}^{-1}$ for these same species. The discrepancy arises partly from the low weighting coefficients that are assigned to extreme response rates in maximum likelihood estimation, and partly from the somewhat horizontal distribution of data points in the area of median response (e.g., Fig. 4). Under conditions of temperature and salinity employed, these limiting values correspond to air-saturation percentages of 49–58% (4.0 – $4.75 \text{ mg} \cdot \text{l}^{-1}$) and 62–68% (5.0 – $5.5 \text{ mg} \cdot \text{l}^{-1}$) (Weiss, 1970). 24-h LC_{50} 's under static conditions were quite low in the case of the cynoglossid and gadid, but higher for the perciform species and the congiopodid (Table I).

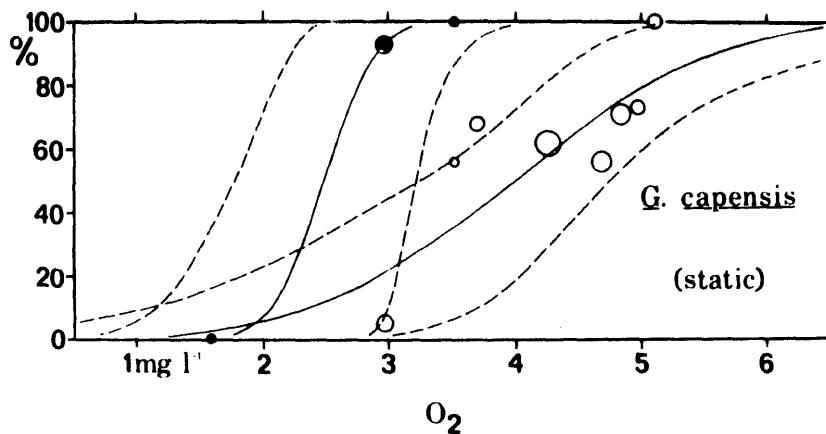


Fig. 4. Response of first-feeding *Gaidropsarus capensis* to low dissolved oxygen under static conditions: other features as in Fig. 1.

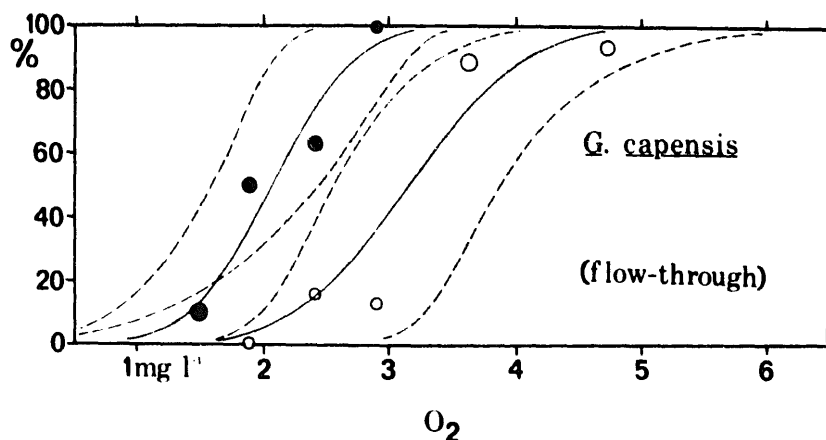


Fig. 5. Response of first-feeding *Gaidropsarus capensis* to low dissolved oxygen under flow-through conditions: other features as in Fig. 1.

Somewhat lower DO concentrations were found to be limiting when the flow-through apparatus was used (Table I, Fig. 5). All the first-feeding larvae tested in this study alternate between states of activity and rest. It is clear that in non-turbulent water, any decline in activity would enhance the oxygen-sink effect, whereas in turbulent water, a decline in activity might be negligible. De Silva & Tytler (1973) showed that herring and plaice larvae lowered their rate of oxygen uptake when the ambient concentration of DO was reduced. The decline in metabolic rate was apparently achieved by a decrease in activity. Since successful first-feeding requires considerable activity, there is a good theoretical basis for the results obtained.

CARBON DIOXIDE

Specific methods

Larvae of one species (*Heteromyceteris capensis*) were exposed to a range of free CO₂ concentrations from 0.46 to 7.7 mg · l⁻¹ CO₂-C. Elevated CO₂ concentrations were achieved by first increasing the alkalinity of system water by adding varying amounts of 0.4 M Na₂CO₃, then reducing the pH to 7.1–7.2 with 0.8 N HCl. It had already been established that, as long as CO₂ is in atmospheric equilibrium, first-feeding in *H. capensis* is not impaired by a 24-h exposure to pH's as low as 6.5, nor by bicarbonate concentrations of that of the most extreme treatment group tested, i.e., 8.9 mmol · l⁻¹.

The tests were static. The test containers were fitted with screened bungs packed with a wad of synthetic wool, turned face-down, and left for 24 h. As usual, food was added and larvae were given 4 h to feed. After rating and removing larvae, the final pH was measured and the total alkalinity was determined by titration with 0.010 N HCl. The nominal concentration of free CO₂ was taken as the amount calculated to be present at the end of the experiment. The relationships and constants required for the evaluation of the carbonate system in sea water were taken from Skirrow (1965).

Results and discussion

There were no noteworthy differences in first-feeding incidence between the control and any of the treatments (Table II). The highest level of CO₂ tested was 7.70 mg · l⁻¹ CO₂-C, which corresponds to 28 mg · l⁻¹ CO₂, or a P_{CO_2} of 1.69 kPa under the conditions specified (Murray & Riley, 1971).

The only study encountered on the tolerance of marine fish eggs of larvae to free CO₂ was that of Kelley (1946), who incubated Pacific herring eggs in the presence of elevated CO₂ (achieved by bubbling CO₂ gas into incoming water). She reported a serious retardation in embryonic development and failure to hatch. Wide variation in the exposure concentration (2.1–18.5 mg · l⁻¹) and pH (5.12–7.88) detract from

TABLE II

Results of an experiment in which first-feeding *Heteromycteris capensis* were exposed to free CO₂ concentrations of 0.46 (control) to 7.70 mg · l⁻¹ CO₂-C.

	pH at end	Total alkalinity (meq · l ⁻¹)	HCO ₃ -C (mg · l ⁻¹)	CO ₃ -C (mg · l ⁻¹)	CO ₂ -C (mg · l ⁻¹)	Number of larvae with				Σ n
						Gut full	Gut ½-full	Gut nearly empty	Gut empty	
Control	7.83	2.45	26.89	1.02	0.46	18	2	0	1	21
Treatment 1	7.09	0.75	8.79	0.06	0.82	12	5	1	1	19
Treatment 2	7.16	3.60	42.41	0.34	3.36	15	4	3	0	22
Treatment 3	7.17	6.00	70.72	0.58	5.48	16	4	0	2	22
Treatment 4	7.20	9.05	106.59	0.94	7.70	14	4	1	1	20

the usefulness of the data. Putnam (1967, unpubl., reviewed by Doudoroff & Shumway, 1970) found that an exposure of rainbow trout eggs during the entire incubation period to 7.7 mg · l⁻¹ CO₂-C had little or no effect on the mass of the newly hatched larvae (minus yolk) as long as dissolved oxygen was adequate.

The Cape sole is admittedly one of the more rugged of the 10 species of larva used in these studies, and one cannot automatically rule out the possibility that first-feeding in one of the other species would have been affected at the highest concentrations tested. Still, 7.7 mg · l⁻¹ is well above the levels of free CO₂ likely to be present in any fish tank. Because excess CO₂ is lost to the atmosphere through the same procedures by which DO is introduced, the chances that free CO₂ could become the primary limiting factor in an adequately oxygenated rearing tank are extremely remote.

GENERAL DISCUSSION

Of the four variables treated here (low and high pH, dissolved oxygen, and carbon dioxide) and the three treated earlier (ammonia, nitrite, and nitrate) (Brownell 1980), only high pH, low dissolved oxygen, and un-ionized ammonia significantly inhibit first-feeding at levels likely to be encountered in a rearing tank for larval marine fish. Although attention is directed to the potential danger of inadequate control of the last three variables, it is clear that other reasons must be found to account for failure of some larvae to feed in captivity. In these studies, ≈20% of all tolerance experiments were rejected because the feeding incidence in the control group was below an arbitrary 50%. Mean control feeding incidence in the remainder was 74.8%. Certain species (*Heteromycteris capensis*, *Gaidropsarus capensis*, *Pachymetopon blochi*, and *Congiopodus spinifer*) only rarely yielded control feeding incidence below 50%; others (*Trulla capensis* and *Synaptura kleini*) were extremely variable often falling below 50% but occasionally approaching 100%. In the bioassay experiments conducted by Lasker (1975), control feeding incidence (defined as the

presence of more than eight particles in the gut at the end of an 8-h exposure to food) ranged from 27 to 32%. About the same percentage had 1–8 particles of food in the gut. The remainder had consumed no food at all. A consumption rate of at least one particle/h had been established in the laboratory as the minimum for survival.

Whether to attribute the often high degree of first-feeding incompetence to factors related to the captive situation, or to blame fundamental disfunction, is a difficult decision at this time. There is tremendous scope for future work on both environmental and genetic factors that influence the ability of a marine fish larva to ingest its first particle of food.

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REFERENCES

- ALDERSON, R. & B. R. HOWELL, 1973. The effect of algae on the water condition in fish rearing tanks in relation to the growth of juvenile sole, *Solea solea* (L.). *Aquaculture*, Vol. 2, pp. 281–288.
- BISHAI, H. M., 1959. The effect of gas content of water on larval and young fish. *Z. wiss. Zool.*, Vol. 163, pp. 37–64.
- BISHAI, H. M., 1960. The effect of hydrogen ion concentration on the survival and distribution of larval and young fish. *Z. wiss. Zool.*, Vol. 164, pp. 107–118.
- BROWNELL, C. L., 1980. Water quality requirements for first-feeding in marine fish larvae. I. Ammonia, nitrite, and nitrate. *J. exp. mar. Biol. Ecol.*, Vol. 44, pp. 269–283.
- DAVIS, J. C., 1975. Minimal dissolved oxygen requirements of aquatic life with emphasis on Canadian species: a review. *J. Fish. Res. Bd Can.*, Vol. 32, pp. 2295–2332.
- DE SILVA, C. D. & P. TYTLER, 1973. The influence of reduced environmental oxygen on the metabolism and survival of herring and plaice larvae. *Neth. J. Sea Res.*, Vol. 7, pp. 345–362.
- DEUBLER, E. E., JR. & G. S. POSNER, 1963. Response of postlarval flounders, *Paralichthys lethostigma*, to water of low oxygen concentrations. *Copeia*, pp. 312–317.
- DOUDOROFF, P. & M. KATZ, 1950. Critical review of literature on the toxicity of industrial wastes and their components to fish. I. Alkalies, acids, and inorganic gases. *Sewage ind. Wastes*, Vol. 22, pp. 1432–1458.
- DOUDOROFF, P. & D. L. SHUMWAY, 1970. Dissolved oxygen requirements of freshwater fishes. *FAO Fish. techn. Pap.*, Vol. 86, pp. 1–291.
- EIFAC (European Inland Fisheries Advisory Commission), 1968. Water quality criteria for European freshwater fish – extreme pH values and inland fisheries. *EIFAC tech. Pap.*, No. 4, 18 pp. (Republished, 1969, *Wat. Res.*, Vol. 3, pp. 593–611.)
- FRY, F. E. J., 1957. The aquatic respiration of fish. In, *The physiology of fishes*, Vol. 1, edited by M. E. Brown, Academic Press, New York, pp. 1–63.
- GRANMO, Å. & S. O. KOLLBERG, 1972. A new simple water flow system for accurate continuous flow tests. *Wat. Res.*, Vol. 6, pp. 1597–1599.
- HIRAYAMA, K., 1970. Studies on water control by filtration system. VI. Acidification of aquarium water. *Bull. Jap. Soc. scient. Fish.*, Vol. 36, pp. 26–34.

- IWAI, T. & G. M. HUGHES, 1977. Preliminary morphometric study on gill development in black sea bream (*Acanthopagrus schlegelii*). *Bull. Jap. Soc. scient. Fish.*, Vol. 43, pp. 929-934.
- JOHANSSON, N. & J. E. KIHLSSTRÖM, 1975. Pikes (*Esox lucius* L.) shown to be affected by low pH values during first weeks after hatching. *Environ. Res.*, Vol. 9, pp. 12-17.
- JONES, J. R. E., 1964. *Fish and river pollution*, Butterworth, Washington, 203 pp.
- JORDAN, D. H. M. & R. LLOYD, 1964. The resistance of rainbow trout (*Salmo gairdnerii* Richardson) and roach (*Rutilus rutilus* (L.)) to alkaline solutions. *Int. J. Air Wat. Pollut.*, Vol. 8, pp. 405-409.
- KELLEY, A. M., 1946. Effect of abnormal carbon dioxide tension on development of herring eggs. *J. Fish. Res. Bd Can.*, Vol. 6, pp. 435-440.
- KILS, U., 1977. The salinity effect on aeration in mariculture. *Meeresforsch.*, Vol. 25, pp. 201-206.
- LASKER, R., 1975. Field criteria for survival of anchovy larvae: the relation between inshore chlorophyll maximum layers and successful first feeding. *Fishery Bull. NOAA*, Vol. 73, pp. 453-462.
- LLOYD, R., 1961. Effect of dissolved-oxygen concentrations on the toxicity of several poisons to rainbow trout (*Salmo gairdnerii* Richardson). *J. exp. Biol.*, Vol. 38, pp. 447-455.
- MILBRINK, G. & N. JOHANSSON, 1975. Some effects of acidification on roe of roach, *Rutilus rutilus* L., and perch, *Perca fluviatilis* L. - with special reference to the Åvaå Lake system in eastern Sweden. *Rep. Inst. freshwat. Res., Drottningholm*, Vol. 54, pp. 52-62.
- MURRAY, C. N. & J. P. RILEY, 1971. The solubility of gases in distilled water and sea water - IV. Carbon dioxide. *Deep-Sea Res.*, Vol. 18, pp. 533-541.
- SAKSENA, V. P. & E. B. JOSEPH, 1972. Dissolved oxygen requirements of newly-hatched larvae of the striped blenny (*Chasmodes bosquianus*), the naked goby (*Gobiosoma bosci*), and the skilfish (*Gobiesox strumosus*). *Chesapeake Sci.*, Vol. 13, pp. 23-28.
- SCHACH, H., 1939. Die künstliche Aufzucht von *C. harengus* L. *Helgoländer wiss. Meeresunters.*, Vol. 1, pp. 359-372.
- SKIRROW, G., 1965. The dissolved gases - carbon dioxide. In *Chemical oceanography*, edited by J. P. Riley & G. Skirrow, Academic Press, London, Vol. 1, pp. 227-322.
- STEWART, N. E., D. L. SHUMWAY & P. DOUDOROFF, 1967. Influence of oxygen concentration on the growth of juvenile large-mouth bass. *J. Fish. Res. Bd Can.*, Vol. 24, pp. 475-494.
- STRUHSAKER, J. W., D. Y. HASHIMOTO, S. M. GIRARD, F. T. PRIOR & T. D. COONEY, 1973. Effect of antibiotics on survival of carangid fish larvae (*Caranx mate*), reared in the laboratory. *Aquaculture*, Vol. 2, pp. 53-88.
- SVERDRUP, H. U., M. W. JOHNSON & R. H. FLEMING, 1942. *The oceans. Their physics, chemistry, and general biology*. Prentice-Hall, New Jersey, U.S.A., 1087 pp.
- SYLVESTER, J. R., C. E. NASH & C. R. EMBERSON, 1975. Salinity and oxygen tolerances of eggs and larvae of Hawaiian striped mullet, *Mugil cephalus* L. *J. Fish Biol.*, Vol. 7, pp. 621-630.
- TROJNAR, J. R., 1977a. Egg and larval survival of white suckers (*Catostomus commersoni*) at low pH. *J. Fish. Res. Bd Can.*, Vol. 34, pp. 262-266.
- TROJNAR, J. R., 1977b. Egg hatchability and tolerance of brook trout (*Salvelinus fontinalis*) fry at low pH. *J. Fish. Res. Bd Can.*, Vol. 34, pp. 574-579.
- WEISS, R. F., 1970. The solubility of nitrogen, oxygen and argon in water and seawater. *Deep-Sea Res.*, Vol. 17, pp. 721-735.
- WICKINS, J. F., 1976. The tolerance of warm-water prawns to recirculated water. *Aquaculture*, Vol. 9, pp. 19-37.